
IOT-BASED SINGLE-AXIS SOLAR TRACKER WITH BATTERY MONITORING AND CLOUD DASHBOARD

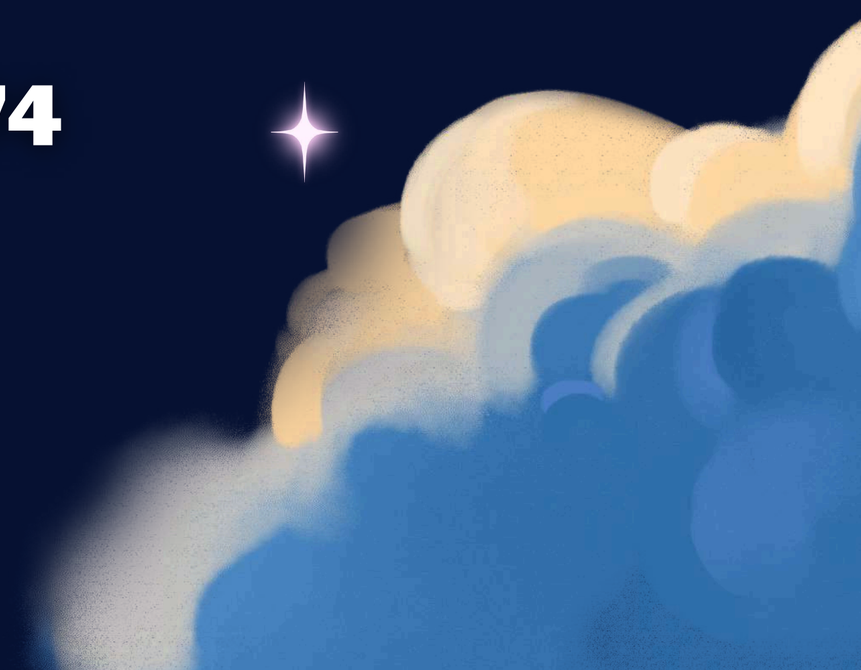
MEMBER

HOÀNG TIẾN MẠNH-SE203727

VÕ ĐẶNG ANH KHOA-SE203865

ĐẶNG THỊ PHƯƠNG ANH-SE203874

NGUYỄN THÀNH LUÂN-SE184179



1. INTRODUCTION

Fixed panel

- Peaks only around solar noon; cosine loss grows quickly in the morning and afternoon.

Sun tracking

- Studies report roughly 20–40% more energy than a fixed panel.

Why single-axis

- One actuator, one hinge and simple control; a good fit for low-power prototypes.

Sensor-driven closed loop: no GPS, real-time clock or site calibration needed.

2. METHOD AND MATERIALS

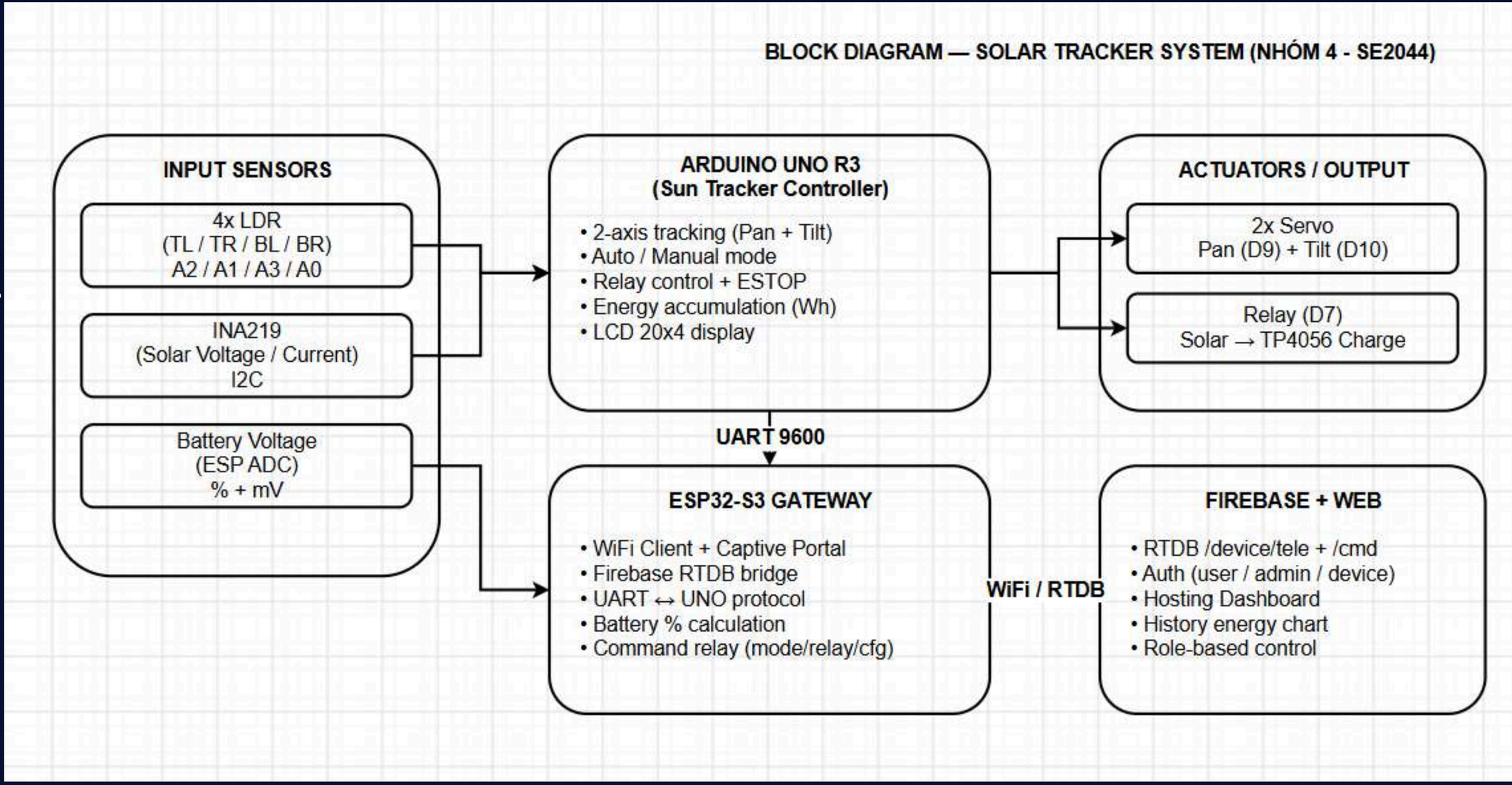


- Microcontroller: ESP32-S3 DevKit N16R8 (integrated Wi-Fi, dual-core processor, 12-bit ADC).
- Light Sensors: Four photoresistor (LDR) sensor modules mounted at the four corners to measure light intensity differences.

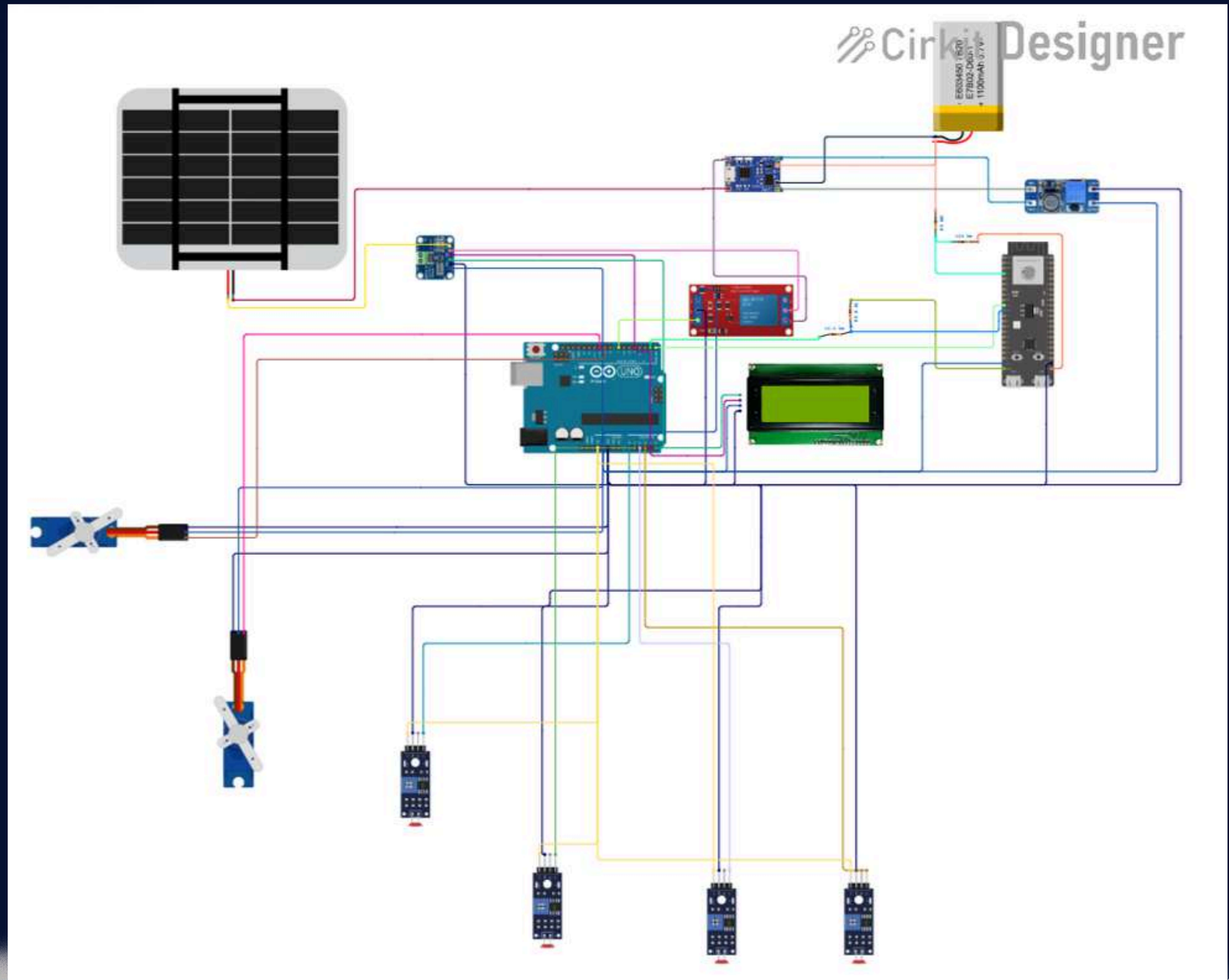
Power System:

- 5W solar panel.
- TP4056 charging module (with battery protection).
- One 18650 Li-ion cell (2500 mAh).
- MT3608 buck converter adjusted to 5V to exclusively supply the servo motor, preventing motor current from causing voltage drops in the microcontroller.
- Micro Servo SG90: Rotation solar panel to where has most lightest .

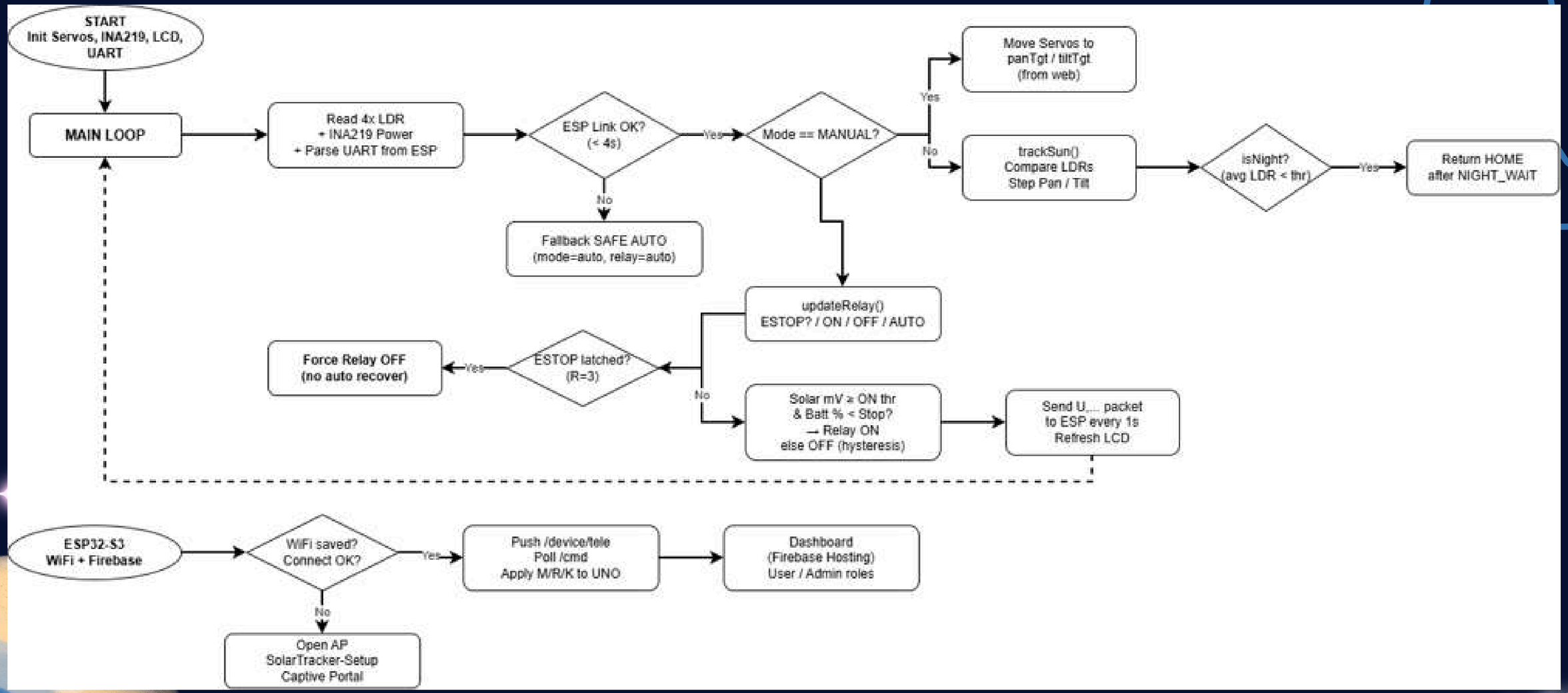
2.1 BLOCK DIAGRAM



3. ELECTRONIC CIRCUIT

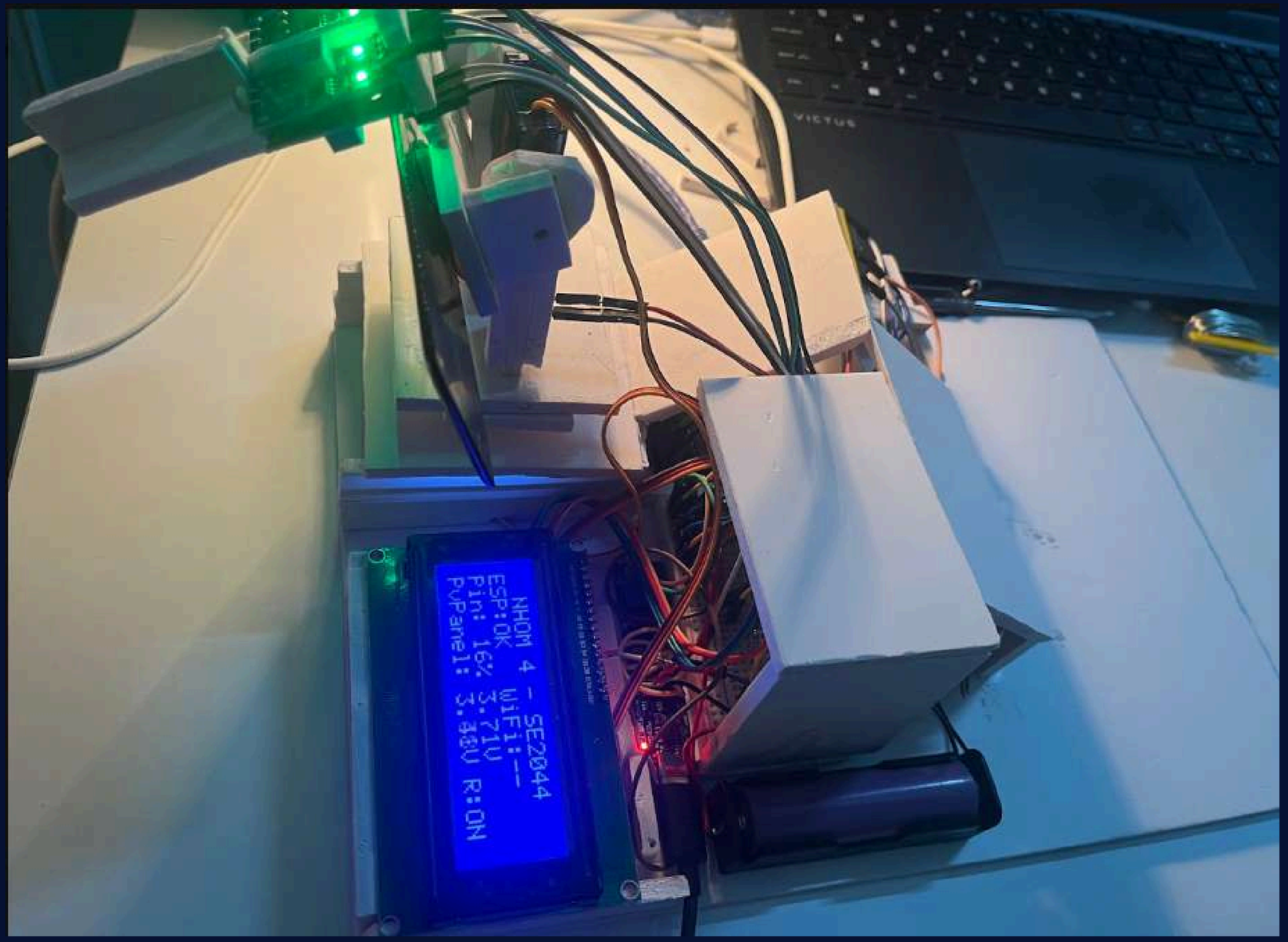


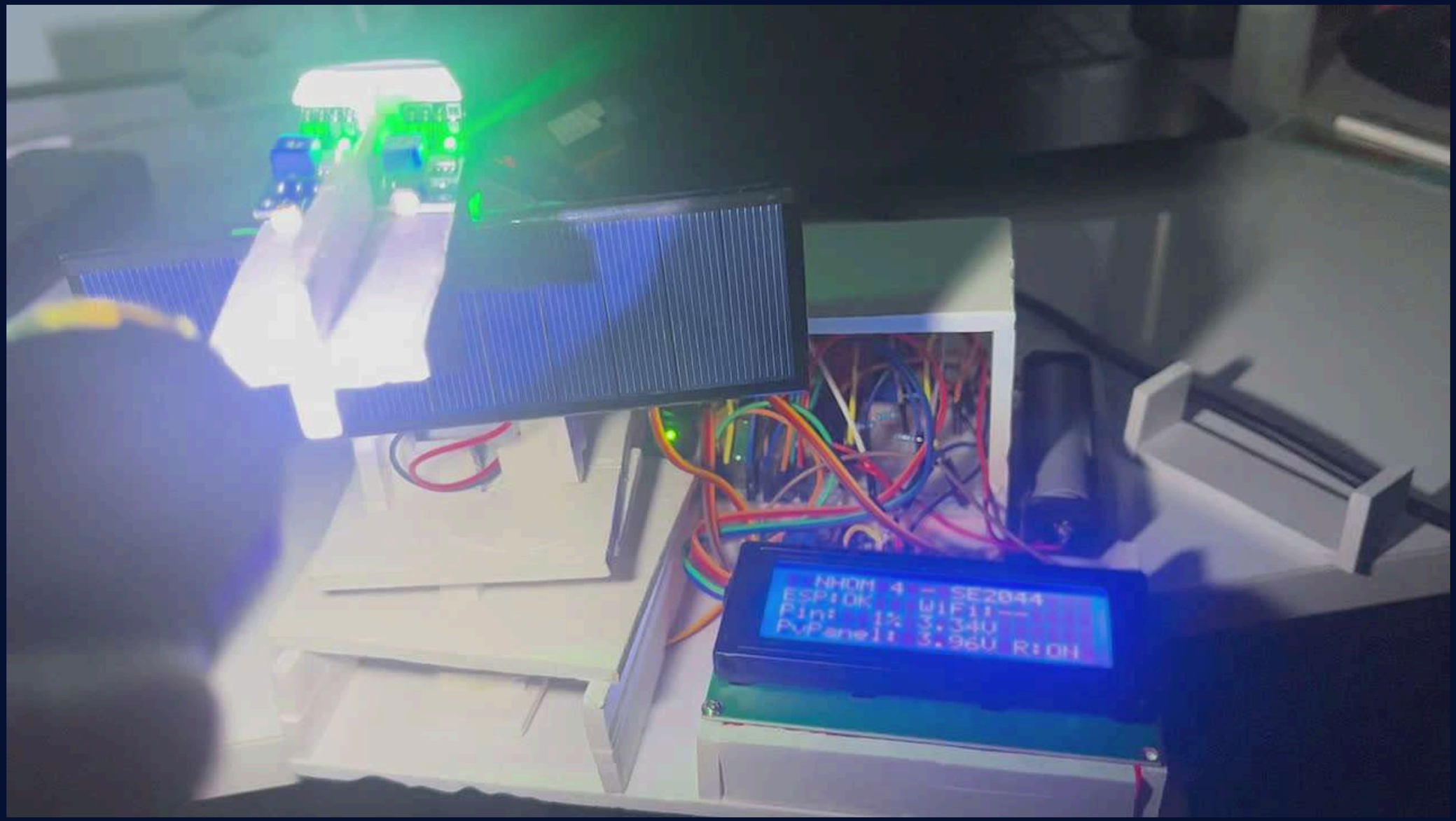
4. FLOWCHART


START
Init Servos, INA219, LCD, UART
MAIN LOOP
Read 4x LDR
+ INA219 Power
+ Parse UART from ESP
ESP Link OK?
(< 4s)
Yes
No
Fallback SAFE AUTO
(mode=auto, relay=auto)
Mode == MANUAL?
Yes
No
Move Servos to
panTgt / tiltTgt
(from web)
trackSun()
Compare LDRs
Step Pan / Tilt
isNight?
(avg LDR < thr)
Yes
Return HOME
after NIGHT_WAIT
updateRelay()
ESTOP? / ON / OFF / AUTO
ESTOP latched?
(R=3)
Yes
Force Relay OFF
(no auto recover)
No
Solar mV ≥ ON thr
& Batt % < Stop?
→ Relay ON
else OFF (hysteresis)
Send U,... packet
to ESP every 1s
Refresh LCD
ESP32-S3
WiFi + Firebase
WiFi saved?
Connect OK?
Yes
No
Open AP
SolarTracker-Setup
Captive Portal
Push /device/tele
Poll /cmd
Apply M/R/K to UNO
Dashboard
(Firebase Hosting)
User / Admin roles

MATERIAL LINK:
HTTPS://DRIVE.GOOGLE.COM/DRIVE/FOLDERS/1N2Y16Y_1SE5SZHCDHP7RT7HPM4M_OQ5O?USP=SHARING

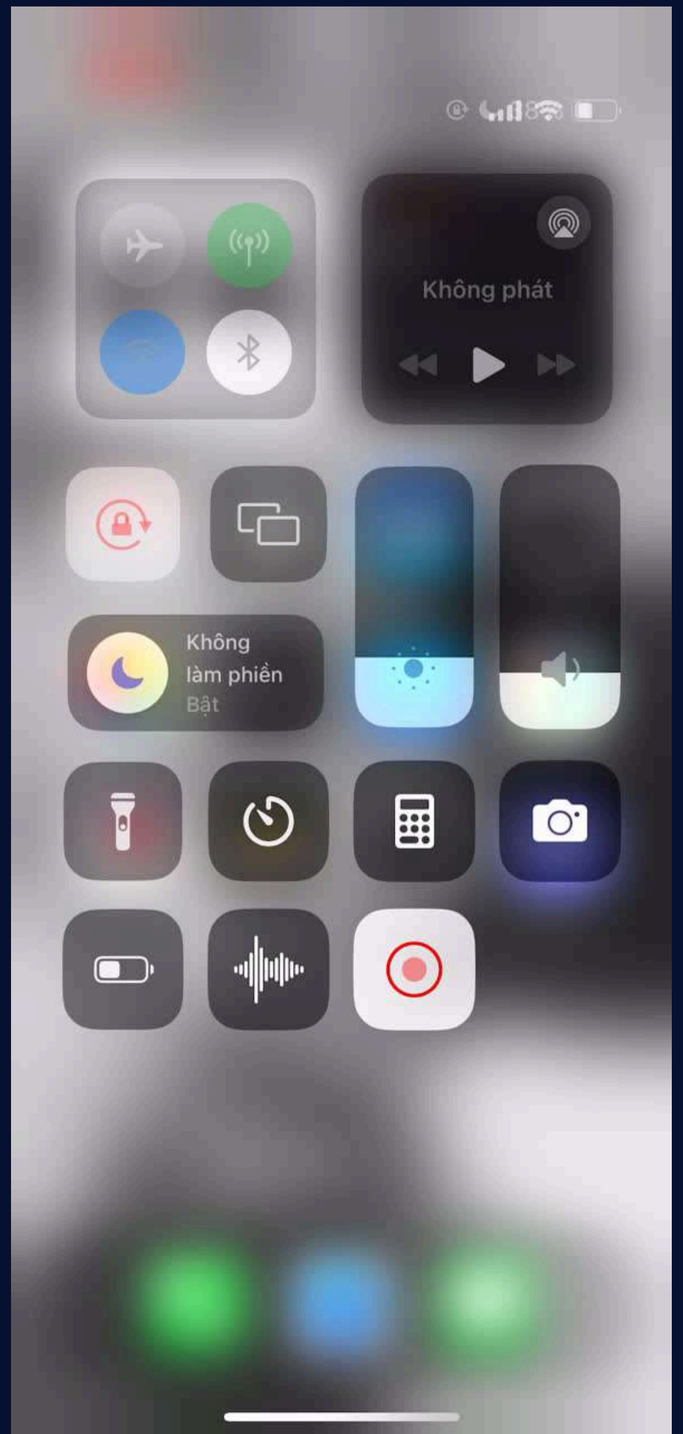
LẮP MẠCH

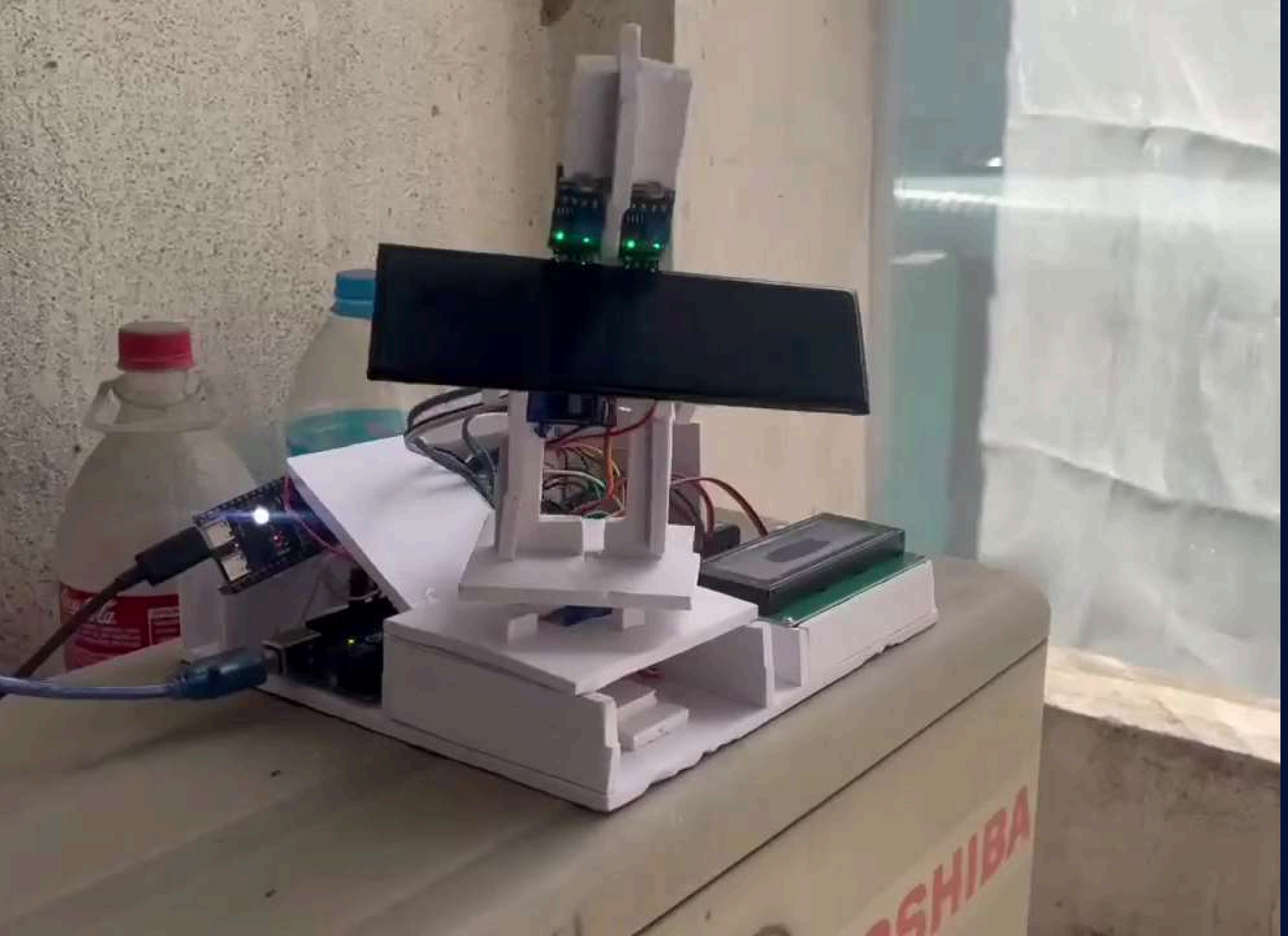




VIDEO DEMO

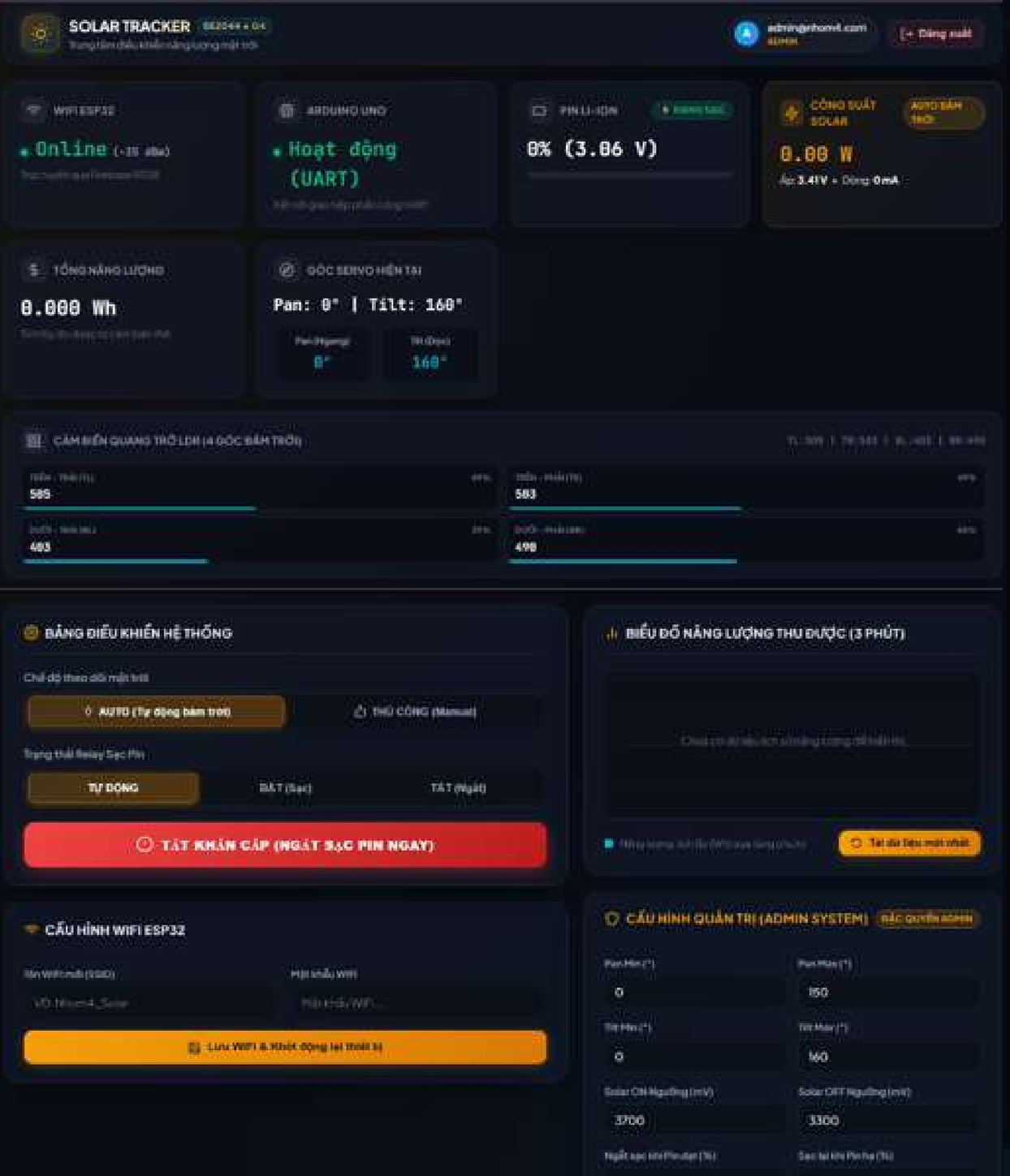
CONFIG WIFI FOR SYSTEM CONNECT INTERNET





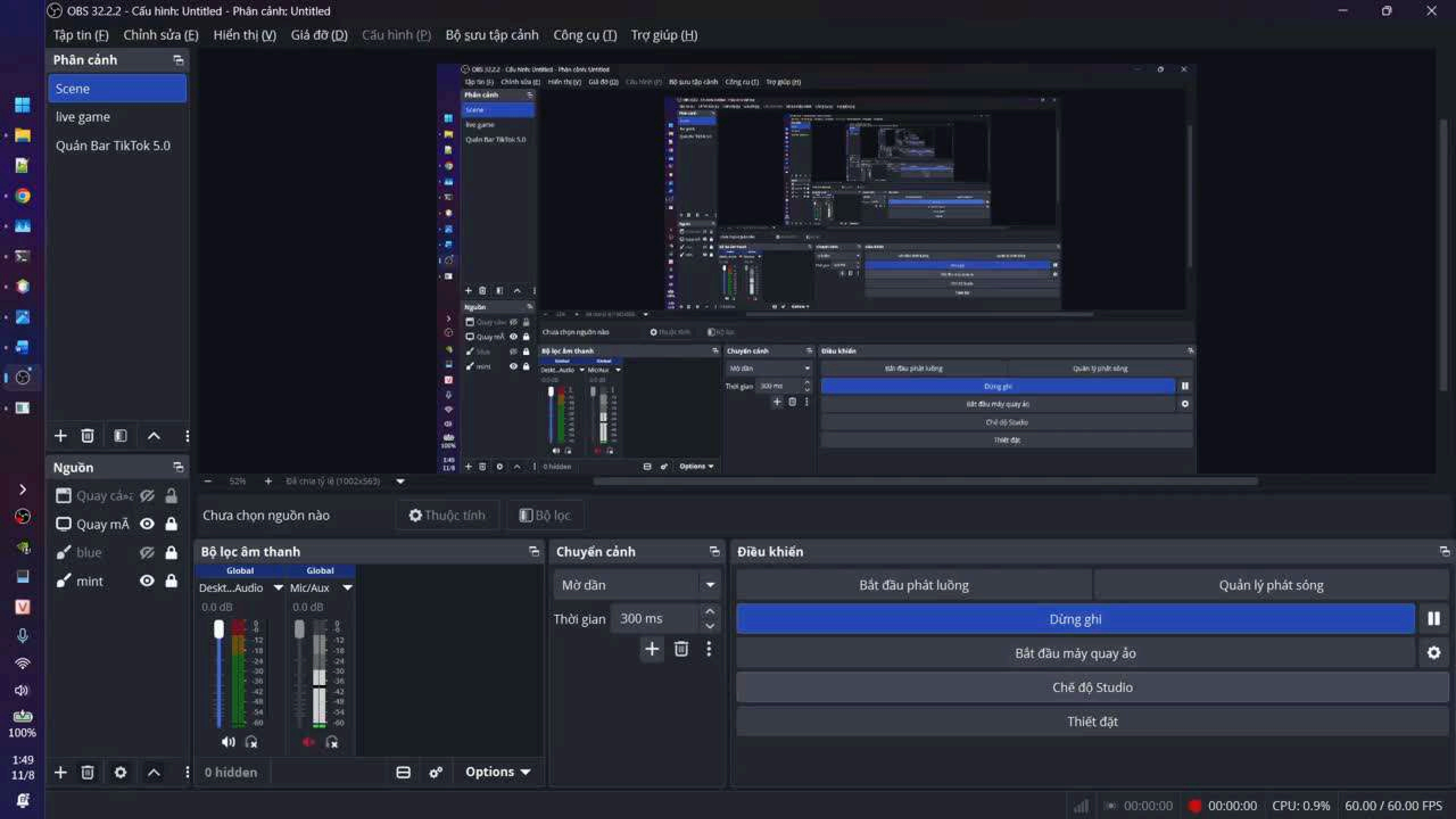
VIDEO DEMO REAL LIFE

DASHBOARD CONTROL





DEMO WEB CONTROLLER


OBS 32.2.2 - Cấu hình: Untitled - Phân cảnh: Untitled
Tập tin (F) Chỉnh sửa (E) Hiển thị (V) Giá đỡ (D) Cấu hình (P) Bộ sưu tập cảnh Công cụ (T) Trợ giúp (H)
Phân cảnh
Scene
live game
Quán Bar TikTok 5.0
Nguồn
Quay cảnc
Quay mã
blue
mint
Chưa chọn nguồn nào
Thuộc tính
Bộ lọc
Bộ lọc âm thanh
Global
Global
Deskt...Audio
Mic/Aux
0.0 dB
0.0 dB
0 hidden
Options
Chuyển cảnh
Mờ dần
Thời gian
300 ms
Điều khiển
Bắt đầu phát luồng
Quản lý phát sóng
Dừng ghi
Bắt đầu máy quay ảo
Chế độ Studio
Thiết đặt
100%
1:49
11/8
00:00:00
00:00:00
CPU: 0.9%
60.00 / 60.00 FPS

5. RESULTS AND CONCLUSION



- Prototype Testing: The system operates stably with smooth rotation, avoiding continuous oscillation thanks to the implemented deadband hysteresis.

- The project has successfully built a low-cost single-axis solar tracking system that optimizes solar energy harvesting, integrates comprehensive hardware/software safety protection mechanisms, and provides online IoT monitoring.

THANK YOU!

